

Demystifying Android App Security

Day 05

25th September 2026

Challenges, Research Directions & Emerging Mobile Security

AGENDA

01



Limitations of Mobile Security Controls

02



Android vs. iOS Security Models

03



Platform-Level Security Research

04



Mobile Malware & Emerging Threats

05



AI-Assisted Mobile Security Analysis

06



Future Directions in Mobile Security

07

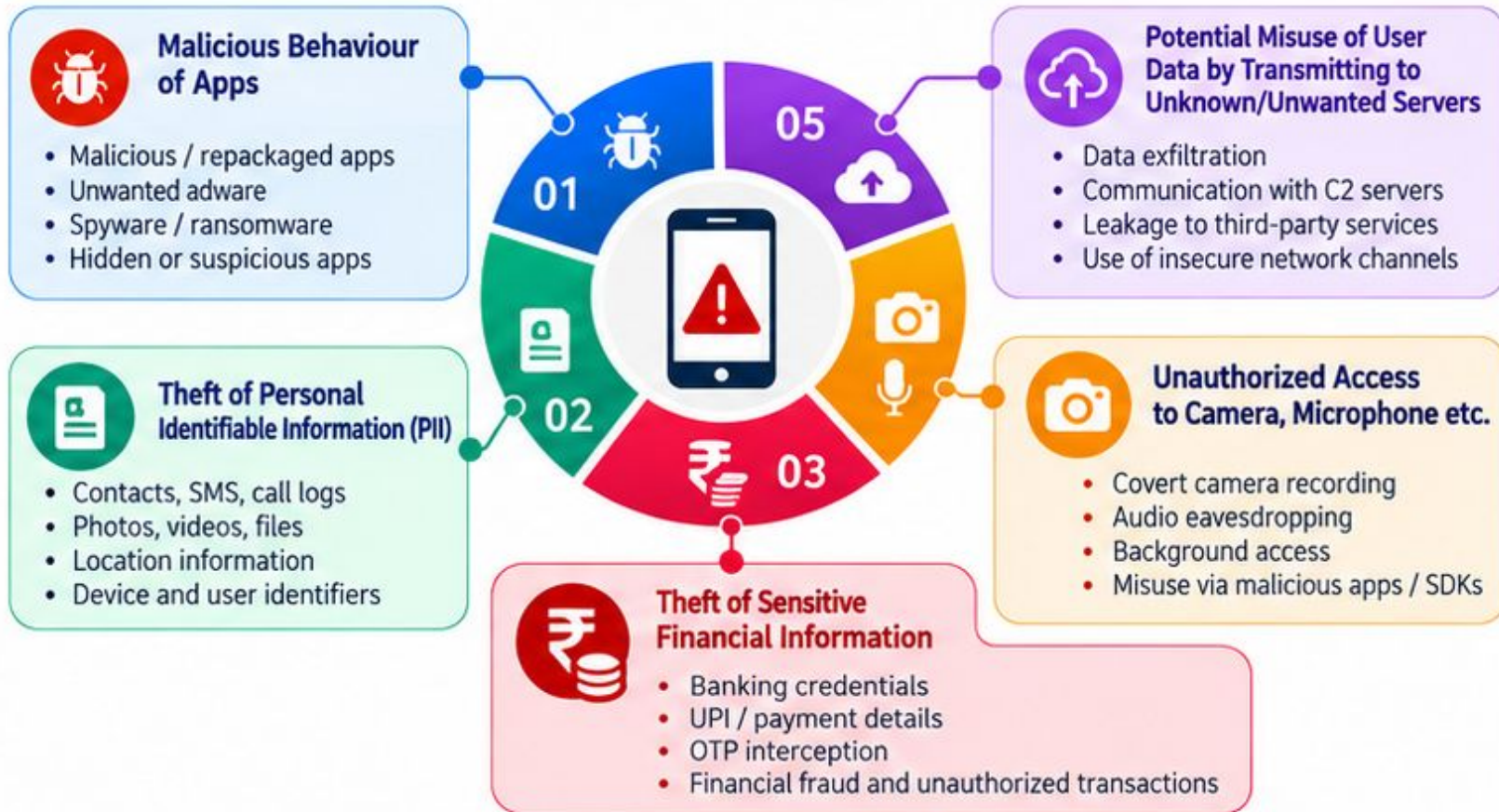


Research Opportunities in AOSP & Mobile Platforms



THREATS

Key Security Threats in Mobile Ecosystem



CHALLENGES

Key Challenges in Mobile Security Analysis and Protection



Manual Analysis Not Scalable

- Time consuming and resource intensive
- Requires skilled analysts
- Not feasible for large scale / real-time analysis

02



Lack of Real-Time Insights of App Behaviour

- Behaviour may not trigger during analysis
- Difficulty in detecting advanced / targeted threats
- Limited visibility into live user environment

04



01

Huge Volume of Apps for Analysis

- Millions of apps in app stores and third-party sources
- Continuous new apps and updates
- Difficult to analyse at scale



03

Minimal Support for Threat Analysis

- Limited visibility into runtime behaviour
- Inadequate threat intelligence for mobile
- Lack of correlation across multiple data sources



Emerging Mobile Threats

1. Application Threats



- Malicious applications
- Repackaged applications
- Supply-chain attacks
- Malicious SDKs
- Trojanized / fake applications

2. Device Threats



- Root / Jailbreak
- Runtime instrumentation
- Privilege escalation
- OS vulnerabilities
- Insecure device configuration

3. Network Threats



- Phishing and smishing
- Malicious URLs
- Man-in-the-Middle attacks
- Command & Control (C2)
- Insecure Wi-Fi / public networks

4. User-Centric Threats



- Social engineering
- Fake applications
- UPI / financial fraud
- Smishing and voice phishing
- Deepfake-enabled attacks



Examples of Malicious Capabilities



Credential
theft



SMS
interception



Overlay
attacks



Device
surveillance



Data
exfiltration



Remote command
execution

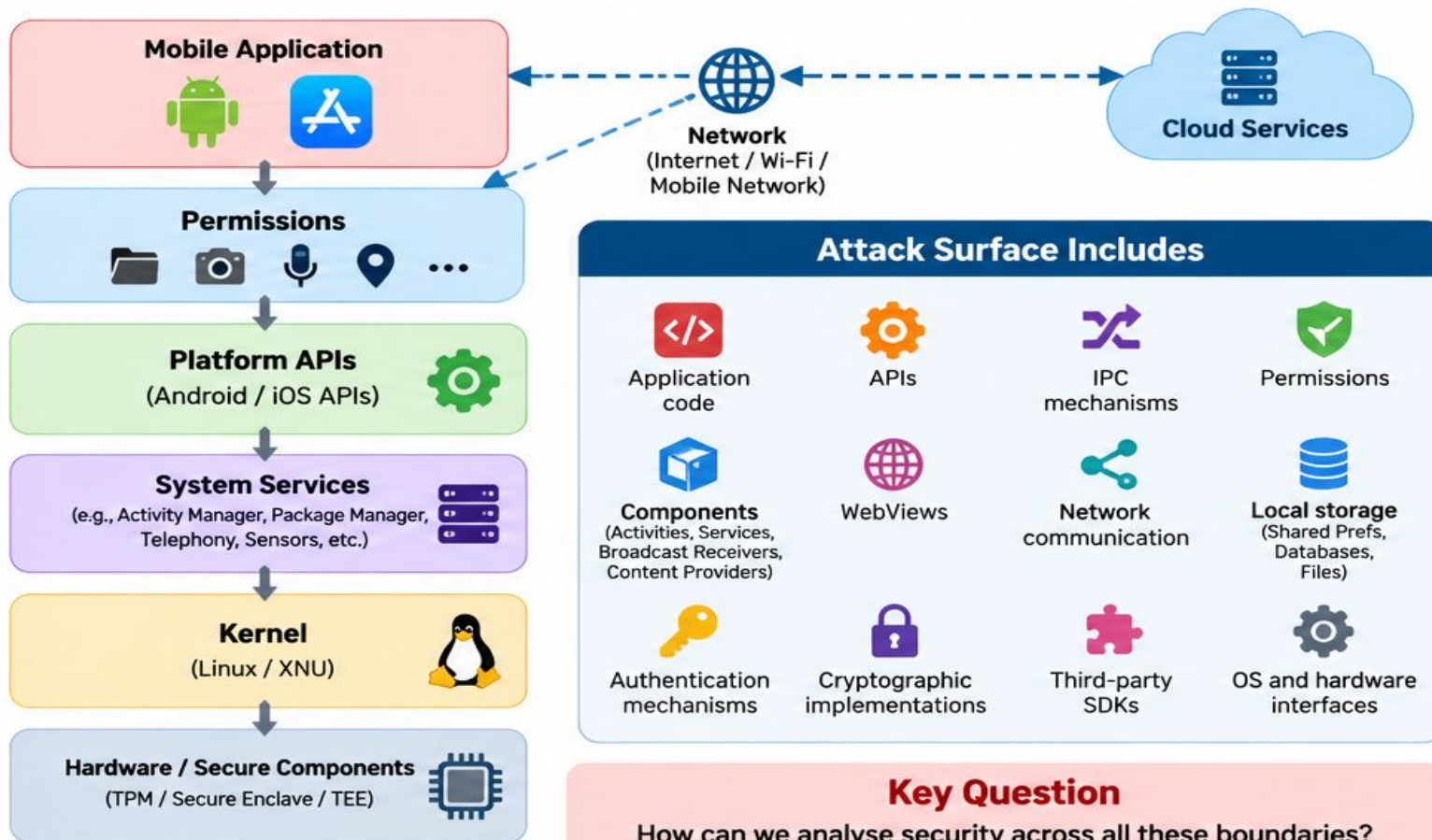


Financial
fraud



C2
communication

Mobile Application Attack Surface



Android vs. iOS Security Models



Android



Ecosystem
Multiple OEMs,
wide device diversity



Source Availability
AOSP available



Application Isolation
Sandbox + SELinux



Application Integrity
Signing + platform controls



Customisation
High (OEM level)



Device Diversity
Very high



Research Surface
OS, OEM, kernel, apps

Key Comparison Areas



Ecosystem



Source Availability



Application Isolation



Application Integrity



Customisation



Device Diversity



Research Surface



iOS



Ecosystem
More controlled
ecosystem



Source Availability
Closed platform



Application Isolation
Sandbox + entitlements



Application Integrity
Strong code-signing
enforcement



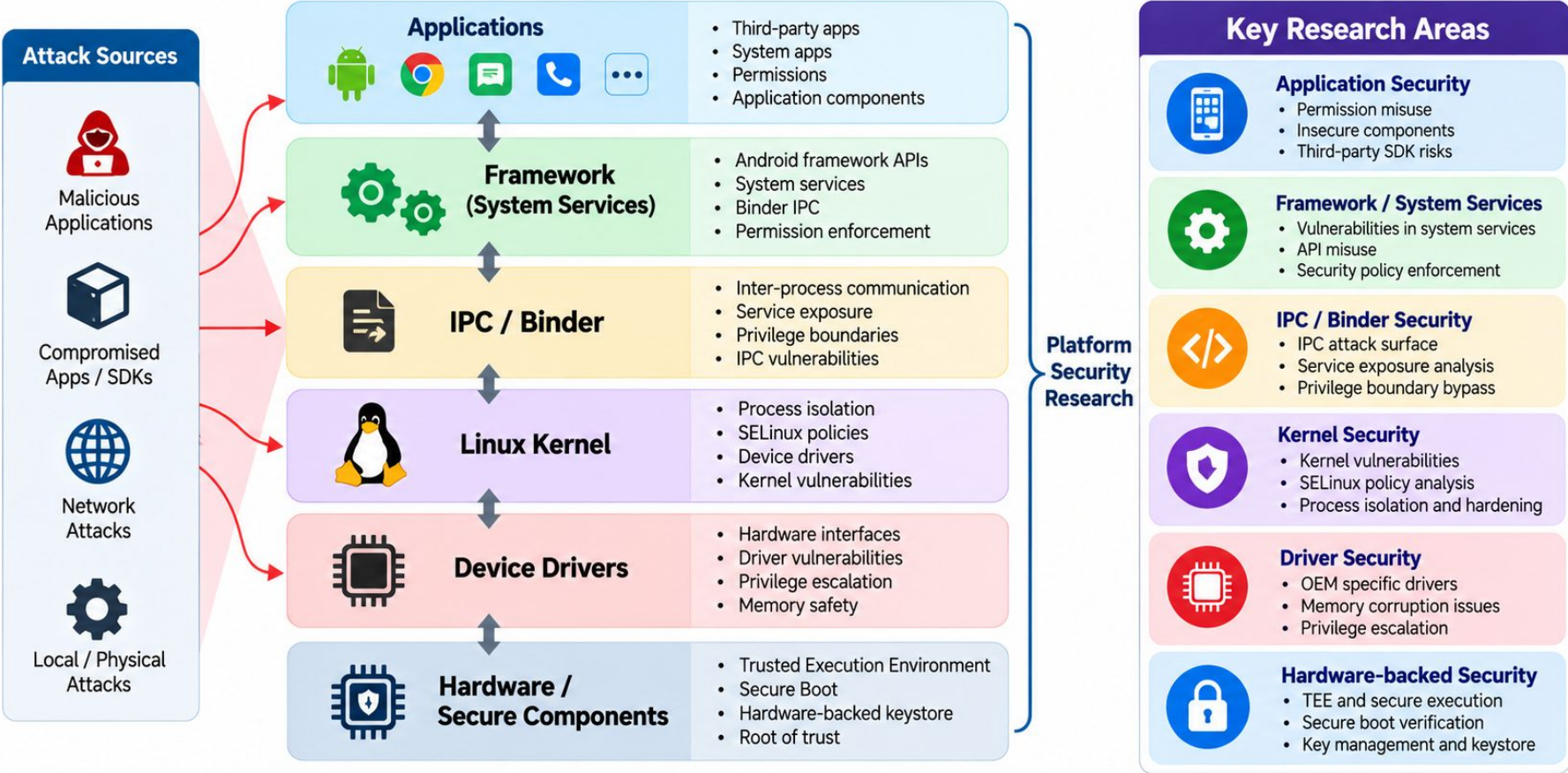
Customisation
Limited



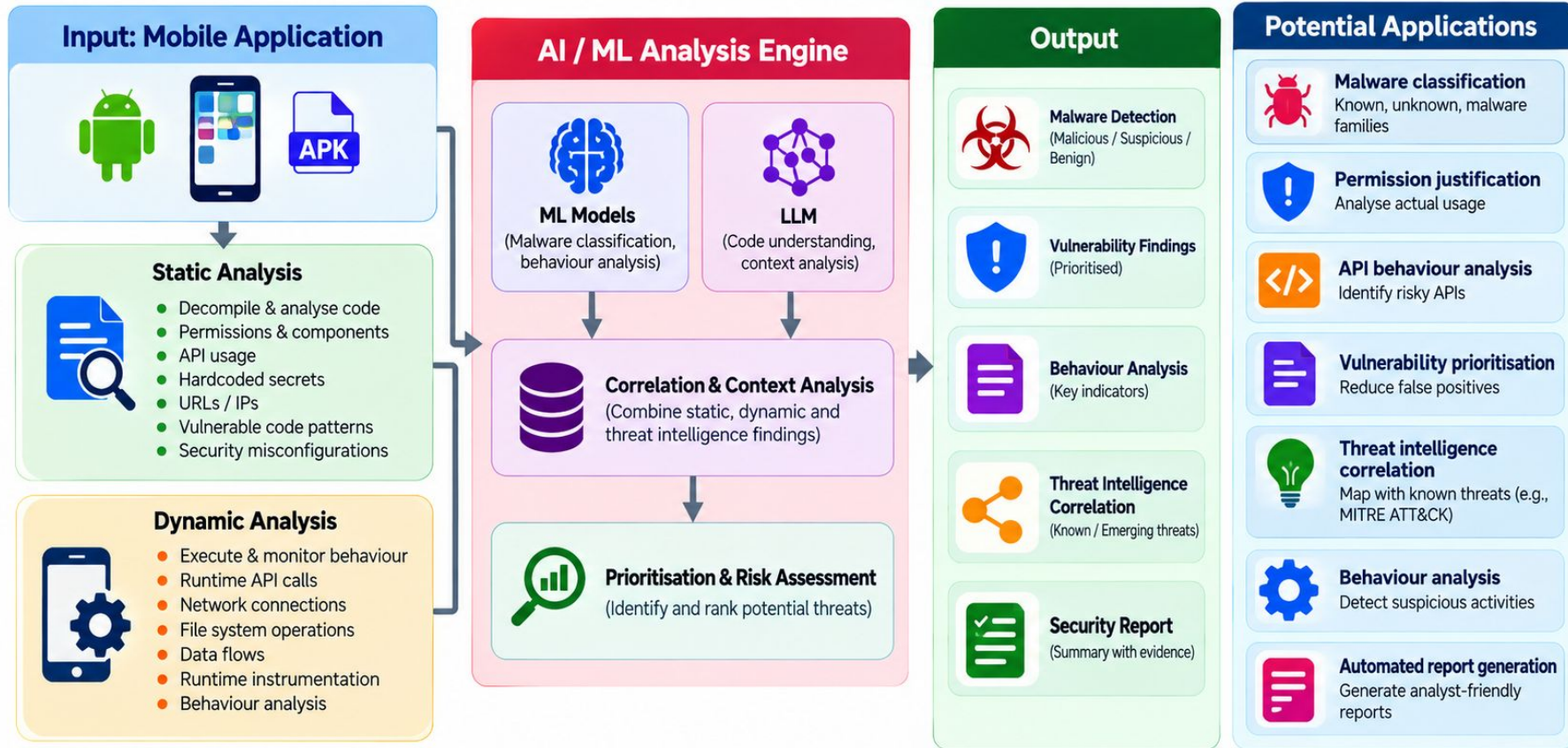
Device Diversity
Comparatively controlled



Research Surface
Apps, runtime,
platform boundaries



AI-Assisted Mobile Security Analysis



AI in Mobile Security: Capabilities and Limitations



Where AI Can Help



Analyse large volumes of security findings



Identify patterns and anomalies



Correlate multiple observations (static, dynamic, threat intelligence)



Explain technical findings (in natural language)



Prioritise potential threats and reduce false positives



Generate security reports and recommendations



Assist security analysts and researchers



Static Analysis Findings



Dynamic Analysis Behaviour



Threat Intelligence Data



Correlated & Prioritised Findings
with Context and Explanation



Analyst Decision

Review, Validate and Take Action



Where AI Can Fail



Hallucinated or incorrect conclusions



Incorrect interpretation of code and behaviour



Insufficient or poor quality evidence



Adversarial inputs and evasion techniques



Model bias and lack of context

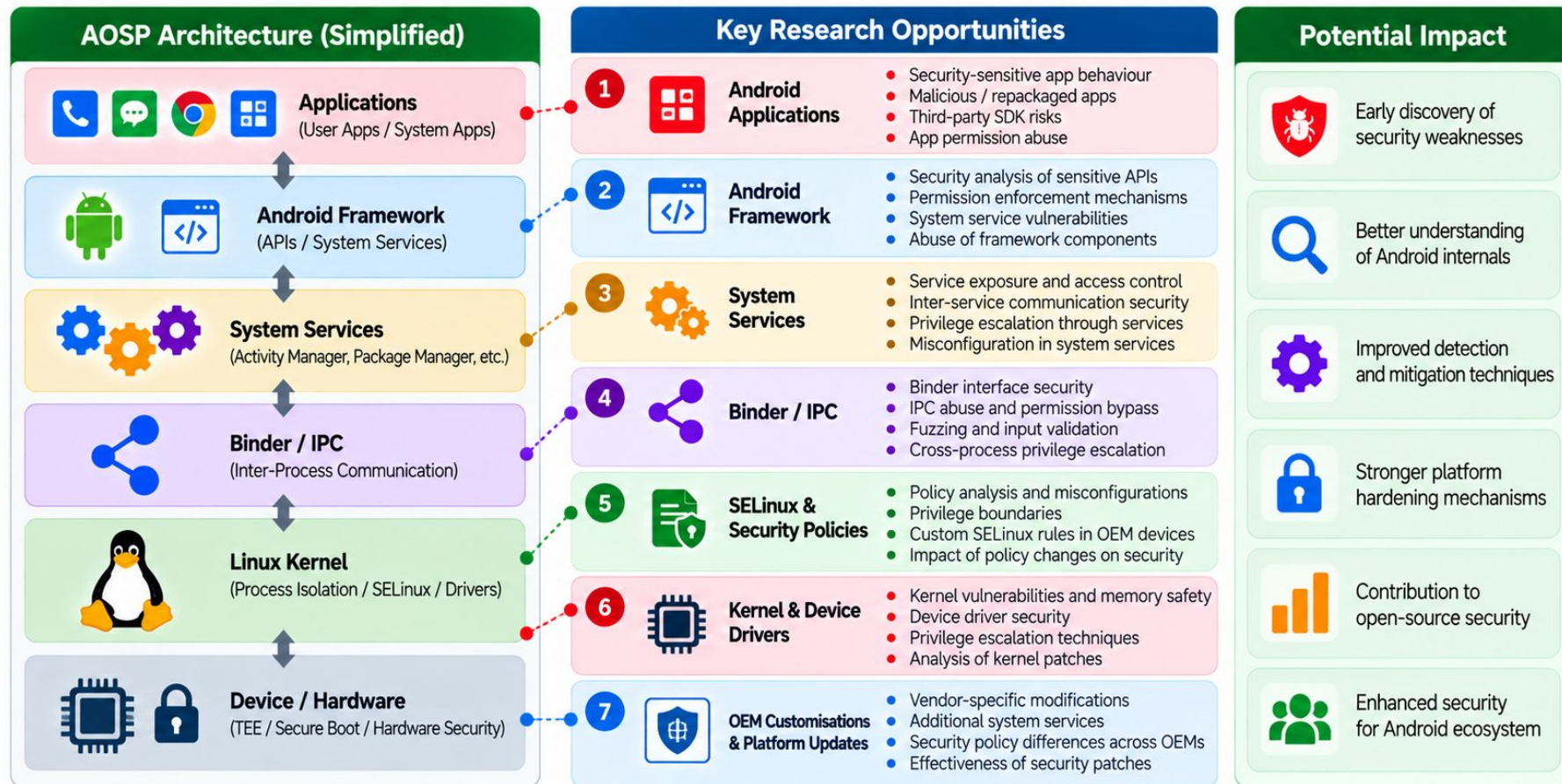


Privacy and data sensitivity concerns



Lack of reproducibility and explainability

Research Opportunities in AOSP (Android Open Source Project)



Contextual Inputs

Device Posture

- Device health & integrity
- Root/Jailbreak detection
- OS version & patch level
- Hardware-backed security

Application Behaviour

- App risk assessment
- Permissions & API usage
- Runtime behaviour monitoring
- Malicious / suspicious activity

User Behaviour

- Usage patterns
- Biometric / behavioural signals
- Anomaly detection
- Continuous authentication

Network Context

- Network type (Wi-Fi / Mobile)
- Suspicious connections
- Malicious domains / URLs
- Location / Geo-fencing

Threat Intelligence

- Known malware indicators
- Emerging threats
- Reputation data
- Real-time threat feeds

Context-Aware Security Engine



On-device
AI / ML



Correlation
& Analytics



Risk Scoring
Engine



Continuous
Risk Assessment

Adaptive Security Response



Allow



Alert



Enforce
Policies



Block /
Remediate

Expected Outcomes



Real-time Threat Detection

Identify malicious and
risky activities



Continuous Risk Assessment

Dynamic risk score
based on context



Adaptive Protection

Appropriate and
context-aware response



Improved User Security & Privacy

Minimal friction with
better protection



Security Insights

Analytics and reports
for continuous
improvement

Key Technologies



Zero Trust

Continuous verification
and least privilege



On-device AI

Privacy-preserving
threat detection



RASP

Runtime application
self-protection



Behavioural Analytics

Device and user
behaviour analysis



Threat Intelligence Integration

Real-time and
emerging threats



Continuous Authentication

Multi-factor and
behavioural signals



Privacy-Preserving Analytics

Security without
exposing sensitive data

Thank You



ISHLESHAK



Vishleshak

Mobile Security & Threat Analysis Platform

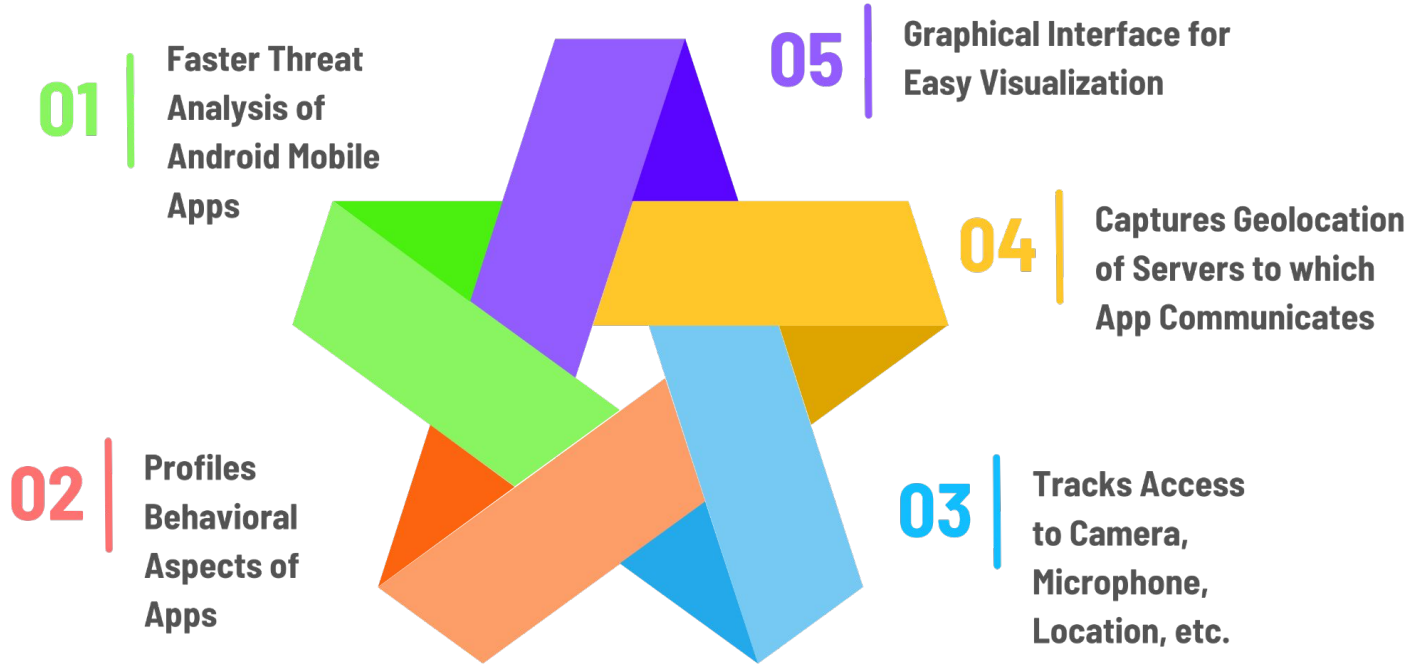
Past one decade has witnessed a multi-fold increase in the volume of malicious apps which seem to have outnumbered the benign apps. The major target of these apps is to misuse the critical resources of the mobile devices, get remote access to the devices & steal critical user data. Hence, there is a need to analyze these apps in order to assess the threats posed by them. But unfortunately, strong Android anti-reversing defenses & code obfuscation are hindering the scope of performing analysis on these apps. This has motivated the team for the development of Vishleshak, an Android based platform to bypass these strong anti reversing defenses and get deep insights into these apps.

Salient Features

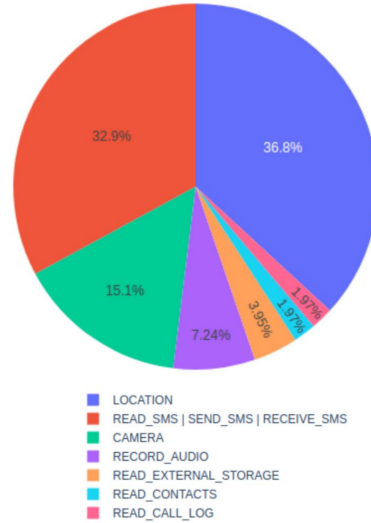
- Evasion of root detection
- Evasion of debuggable properties
- Resource access monitoring
- Supports instrumentation tools
- Monitor app specific network endpoint communications
- Tracking usage of dangerous permissions



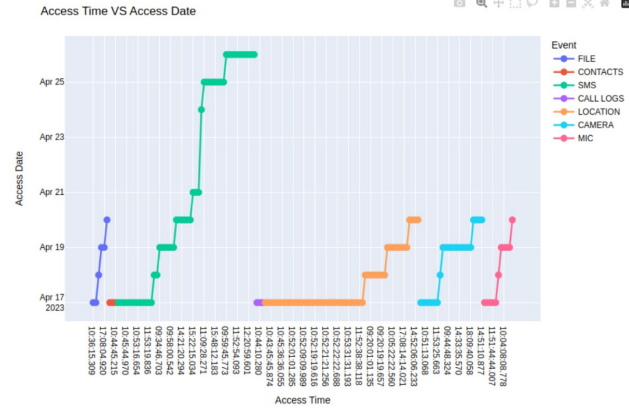
Features



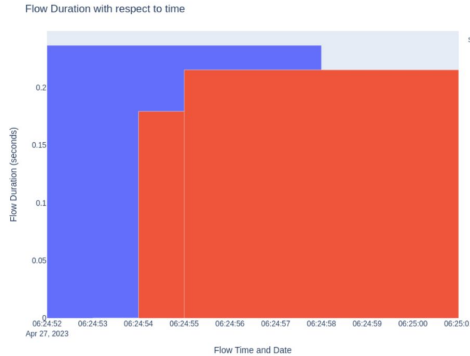
Dangerous Permission Access Analysis:



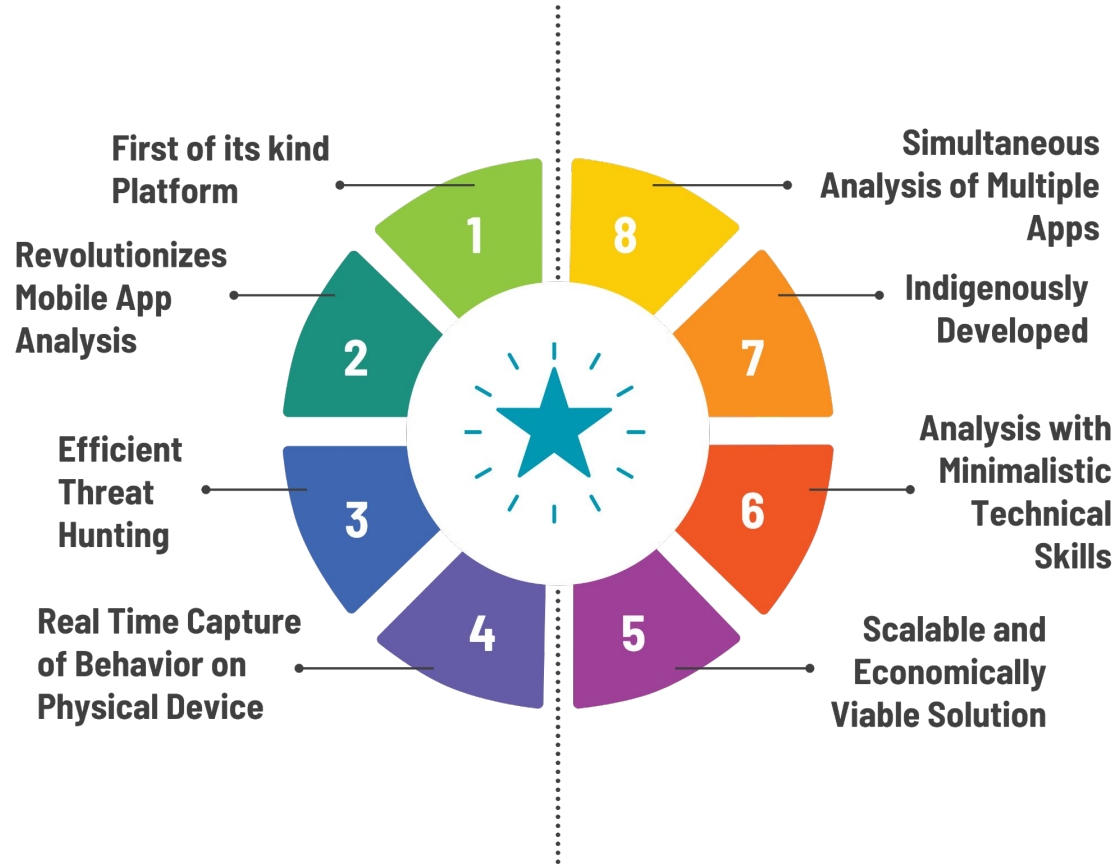
App Profiling:



Network Endpoint Monitoring:



USPs



Parikshan

Automated Security Analysis of Mobile Apps



PARIKSHAN

Insightful App Security Analysis

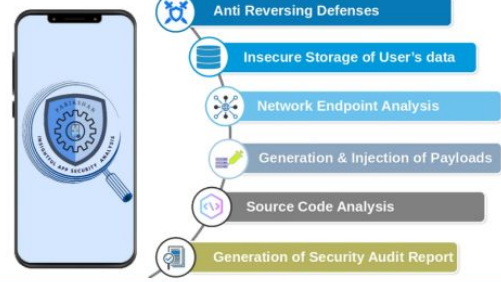
Mobile applications witnessed rampant proliferation from multiple App Stores and these applications handles user's sensitive information and critical device resources. Hence, the security assessment & penetration testing of the mobile applications has become paramount in-order to prevent mis-utilization of data & resources. Apparently, an automation tool performing fast pace security analysis becomes inevitable. Parikshan is an Automation tool mainly focused on performing static and dynamic analysis of mobile applications. The tool has the capabilities to identify the security vulnerabilities and perform penetration testing for few of them. As an outcome, a detailed security audit report is generated containing the information about the identified vulnerabilities which aids to carry out further analysis.

Salient Features:

- Automated Security Analysis, Vulnerability Assessment & Penetration Testing of Mobile Applications.
- Adheres to OWASP Top 10 Mobile Vulnerabilities and executes a total of 24 Static & 12 Dynamic test cases.
- Analysis of device logs and storage to identify leakage of user sensitive data.
- Generation and Injection of payloads during Dynamic Analysis.
- Real time capture of Geo-location information of the network endpoints communicated by the application
- Generation of detailed Audit Report of identified Vulnerabilities

Target Audience :

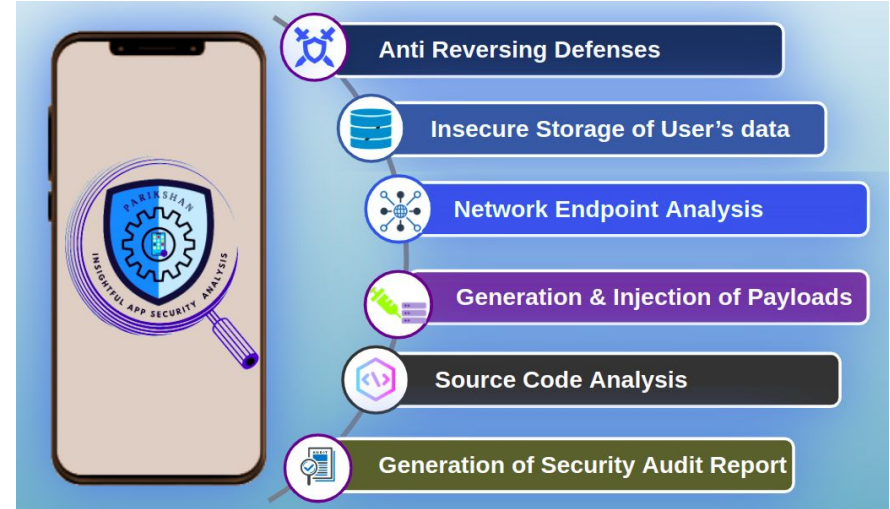
- Mobile Application Developers
- Mobile Application Security Analysts



प्रगत संगणन विकास केंद्र
CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

Parikshan

- ✓ PARIKSHAN is an Automation tool which aids in performing Static & Dynamic Security Analysis of the Android-based Mobile Applications.
- ✓ Supports:
 - Automated Security Analysis, Vulnerability Assessment, and Penetration Testing
 - Android based Mobile Applications
- ✓ Use case for the specified sector:
 - Aids Security Analysts, Developers & Security Audit Labs in performing Security Analysis & Vulnerability Assessment of the Mobile Applications



Parikshan - User Interface

Email address

Password

[Forgot password?](#)

PARIKSHAN

You can input ".apk" file

Apk File: No file chosen

Please select which analysis you want to perform

☒ Static Analysis

☐ Dynamic Analysis

Static Analysis Results

Feature	Value
Application Name	Parikshan Demo App
Application Version	1.0
MD5 Hash	52b7e1c2664f4c21a4d6cfef92cc8403
Main Activity	org.cdac.insecurerregistrationapp.MainActivity
Target SDK	33
Min SDK	21
Package Name	org.cdac.parikshandemo

Certificate Details

Name	Details
v1 signature scheme	true
v2 signature scheme	true
v3 signature scheme	false
Public Key SHA1	26e7d99d375eacfc6d40599ef76ea2757059f
Public Key SHA256	7b626c015b73e653d1e3d408ad412c57ba804304e35a510d5917fa0250dab06
Key algorithm	RSA
MD5	ae1a75adb16db324b6a5bfe06771134d5
Key Algorithm used	RSA

Automated Dynamic Analysis Testing

Below are the analysis checklist which are included under 'Automated Dynamic Analysis'

M- Prabandh

Enterprise Mobile Device Management



M-PRABANDH Mobile Device Management

In recent times, mobile devices are not only used for personal purposes but also increasingly employed for managing businesses due to their convenience and mobility. This has led to a significant amount of sensitive business information being stored on these devices, making security a crucial responsibility for employers. This is where Mobile Device Management (MDM) is essential for securing organizational data in today's mobile-driven business landscape. This has motivated C-DAC to develop an M-Prabandh MDM solution that allows organizations to effectively monitor & manage mobile devices and reduce the risk of data breaches & theft. It also aids in safeguarding business critical information, and uphold data security standards.

The centralized management dashboard allows organizations to take control of the mobile devices effortlessly and provide a comprehensive interface, seamlessly presenting information about all connected devices. The solution supports a hassle-free device onboarding, efficient policy enforcement, and user device management. It also helps in gaining valuable insights into device specifications, real-time statistics, and get comprehensive information for informed decision-making. Further, features such as the centralized security policy management, app risk assessment, and role-based access control secures the devices and provides controlled access to the business data.

Saillant Features



1. Centralized Management Dashboard
2. Seamless Device Onboarding
3. MDM Agent for Enhanced Control
4. Role-based Access Control
5. App Risk Assessment & Monitoring
6. App Installation Tracking & Management
7. App Usage Monitoring & Analytics
8. Real-time Location Tracking of Devices

प्रगत संगणन विकास केंद्र
CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

M-Prabandh MDM solution allows organizations to effectively monitor & manage mobile devices and reduce the risk of data breaches & theft. It also aids in safeguarding business critical information, and uphold data security standards





← → ↻ ⚠ Not secure 10.244.2.156:4200/dashboard

M-PRABANDH

⌵

Home

Enroll Devices

Device Management

APK Analysis

Geo-Tracking

Dashboard

5

Enrolled Users

4

Enrolled Devices

0

Connected Devices

Device locations

Map ☒ Globe

All Active & Inactive users

INACTIVE U...

ACTIVE USE...

Legend

ACTIVE USERS

INACTIVE USERS

TOTAL A...

ACTIVITY LOGS

Actions

More

Select an option

LOGGED-IN

2/9/24, 9:38 AM

Device

securemdm@gmail.com

Logged-in.

APP INSTALLED (Android System WebView)

2/8/24, 3:51 PM

Device

mkavachi@cdac.in

App installed.

APP UNINSTALLED (c)

2/8/24, 3:51 PM

Device

mkavachi@cdac.in

App uninstalled.

APP INSTALLED (SecureFamily)

2/8/24, 12:28 PM

Device

mkavachi@cdac.in

App installed.

APP UNINSTALLED (c)

2/8/24, 12:28 PM

Device

mkavachi@cdac.in

App uninstalled.

APP INSTALLED (SOTI MobiControl)

2/7/24, 3:19 PM

Device

mkavachi@cdac.in

App installed.

APP INSTALLED (SecureFamily)

2/7/24, 12:36 PM

Device

mkavachi@cdac.in

App installed.

APP UNINSTALLED (c)

2/7/24, 12:36 PM

Device

mkavachi@cdac.in

App uninstalled.



- Home
- Enroll Devices
- Device Management
- APK Analysis
- Geo-Tracking

Device information

Android
Platform Type

6%
Battery percentage

Device storage

View History

Summary Apps Installed Actions Disconnected

Total applications (181)

ALL APPLICATIONS Search Applications

App Name	Packge name	Package ID	Risk level	Actions
AFK Arena	com.lilithgames.hgame.gp.id	1.59.03	BENIGN	
AJIO	com.ril.ajio	9.2.0	BENIGN	
AlipayHK	hk.alipay.wallet	3.15.1.1	MALIGN	
aha	ahaflix.tv	4.0.19	BENIGN	
adware	com.example.adware	1.0.0	BENIGN	
13Card Rummy	com.Indianrummy.onlinenummy.cardrummy	1.4	BENIGN	
Airtel Xstream	tv.accedo.airtel.wynk	1.63.1	BENIGN	
AirMirror	com.sand.airmirror	1.1.3.0	BENIGN	

Showing 1 to 8 entries 1 2 3 4 5

8:01

App Info

AFK Arena
com.lilithgames.hgame.gp.id

AJIO
com.ril.ajio

AlipayHK
hk.alipay.wallet

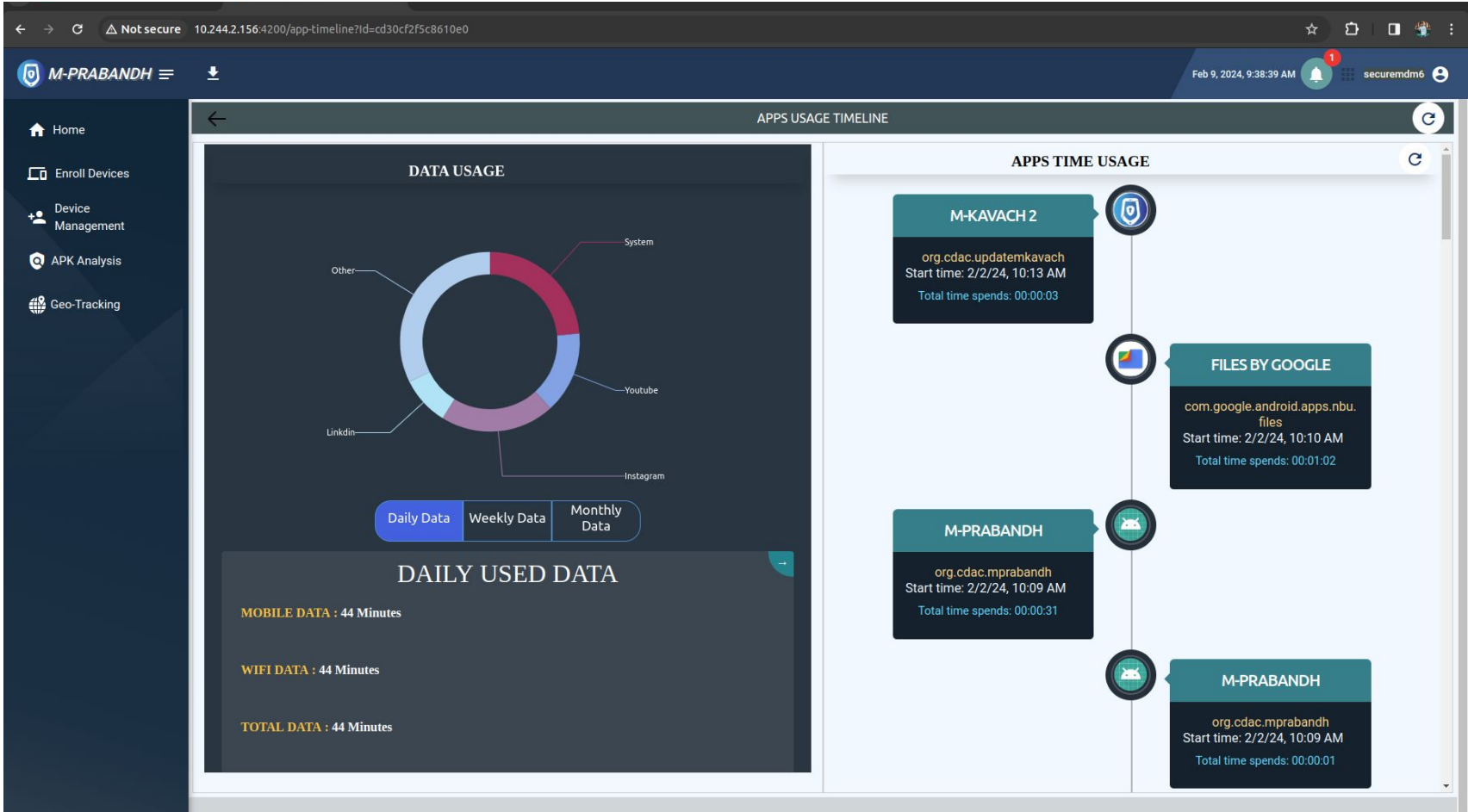
aha
ahaflix.tv

adware
com.example.adware

13Card Rummy
com.Indianrummy.onlinenummy.cardrummy

Airtel Xstream
tv.accedo.airtel.wynk

AirMirror
com.sand.airmirror



M-Kavach 2



M-KAVACH 2

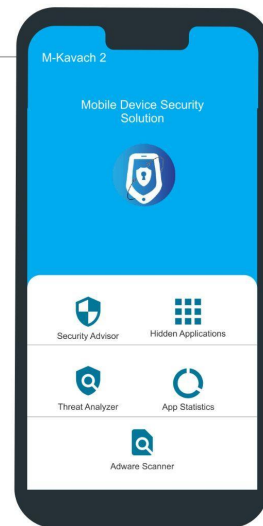
M-Kavach 2 is a comprehensive mobile device security solution addressing emerging threats related to Android based mobile devices. The major emphasis is on advising the users against security misconfigurations, detection of hidden/ banned apps and scanning the device for potential malicious apps installed on the user's mobile device.



Scan & Install

Salient Features

- **Threat Analyzer:**
Detects potential malicious apps on the user's device.
- **Security Advisor:**
Provides a holistic security status of the device.
- **Detection of Hidden/Banned Apps:**
Identifies the existence of any hidden/banned apps on the user's device.
- **App Latest Update Statistics:**
Notify the users regarding the apps not updated for longer durations.
- **Adware Scanner:**
Notify the users about the adware present on the device.



Contact Details :

✉ mkavach@cdac.in

☎ +91 90142 02885



M-Kavach 2

M-Kavach 2 is a comprehensive mobile device security solution addressing emerging threats related to Android based mobile devices. The major emphasis is on advising the users against security misconfigurations, detection of hidden/banned apps and scanning the device for potential malicious apps installed on the user's mobile device.

Salient Features

